Lake Powell and Lake Mead Historical and Current Reservoir Data

David E. Rosenberg

This report was generated on: September 27 , 2026

Overview

This is an R Markdown document. The document generates plots for Lake Powell and Lake Mead:

1. Annual Lake Powell Release Compared to 7.5 and 8.23 Targets

Data wrangling strategy

1. Import the reservoir elevation-volume curves.
2. Import Daily, Monthly, and Annual Lake Powell and Lake Mead data from the Reclamation's HydroData Web portal https://www.usbr.gov/uc/water/hydrodata/reservoir_data/site_map.html
3. Filter the reservoir data for the appropriate variables/fields.

This is a beginning R-programming effort! There could be lurking bugs or basic coding errors that I am not even aware of. Please report bugs/feedback to me (contact info below)

Requested Citation

David E. Rosenberg (2026), "Lake Powell and Lake Mead Historical and Current Reservoir Data." Utah State University. Logan, Utah. https://github.com/dzeke/ColoradoRiverCollaborate/tree/main/HistoricalCurrentReservoirData.

```
## [1] "C:/Rosenberg/Work/USU/Research/ColoradoRiver/RCode/ColoradoRiverCollaborate/HistoricalCurrentRes
## [1] "C:/Rosenberg/Work/USU/Research/ColoradoRiver/RCode/ColoradoRiverCollaborate/AutoReadUSBRData/df
## [1] "C:/Rosenberg/Work/USU/Research/ColoradoRiver/RCode/ColoradoRiverCollaborate/AutoReadUSBRData/df
## [1] "C:/Rosenberg/Work/USU/Research/ColoradoRiver/RCode/ColoradoRiverCollaborate/AutoReadUSBRData/df
## [1] "Successfully read - Lake Powell : Storage  -  https://www.usbr.gov/uc/water/hydrodata/reservoir
## [1] "Successfully read - Lake Powell : Evaporation  -  https://www.usbr.gov/uc/water/hydrodata/reserv
## [1] "Successfully read - Lake Powell : Inflow  -  https://www.usbr.gov/uc/water/hydrodata/reservoir_
## [1] "Successfully read - Lake Powell : Inflow Volume  -  https://www.usbr.gov/uc/water/hydrodata/res
## [1] "Successfully read - Lake Powell : Unregulated Inflow  -  https://www.usbr.gov/uc/water/hydrodata
## [1] "Successfully read - Lake Powell : Unregulated Inflow Volume  -  https://www.usbr.gov/uc/water/hy
## [1] "Successfully read - Lake Powell : Total Release  -  https://www.usbr.gov/uc/water/hydrodata/res
## [1] "Successfully read - Lake Powell : Release volume  -  https://www.usbr.gov/uc/water/hydrodata/re
## [1] "Successfully read - Lake Powell : Pool Elevation  -  https://www.usbr.gov/uc/water/hydrodata/re
## [1] "Successfully read - Lake Powell : Area  -  https://www.usbr.gov/uc/water/hydrodata/reservoir_da
## [1] "Successfully read - Lake Mead : Storage  -  https://www.usbr.gov/uc/water/hydrodata/reservoir_d

## Error in file(file, "rt") : cannot open the connection
## [1] "Skipping due to error - Lake Mead : Evaporation  -  https://www.usbr.gov/uc/water/hydrodata/res

## Error in file(file, "rt") : cannot open the connection
## [1] "Skipping due to error - Lake Mead : Inflow  -  https://www.usbr.gov/uc/water/hydrodata/reservoi

## Error in file(file, "rt") : cannot open the connection
## [1] "Skipping due to error - Lake Mead : Inflow Volume  -  https://www.usbr.gov/uc/water/hydrodata/r
```

```
## Error in file(file, "rt") : cannot open the connection
## [1] "Skipping due to error - Lake Mead : Unregulated Inflow  -  https://www.usbr.gov/uc/water/hydroda

## Error in file(file, "rt") : cannot open the connection
## [1] "Skipping due to error - Lake Mead : Unregulated Inflow Volume  -  https://www.usbr.gov/uc/water/
## [1] "Successfully read - Lake Mead : Total Release  -  https://www.usbr.gov/uc/water/hydrodata/reserv
## [1] "Successfully read - Lake Mead : Release volume  -  https://www.usbr.gov/uc/water/hydrodata/reser
## [1] "Successfully read - Lake Mead : Pool Elevation  -  https://www.usbr.gov/uc/water/hydrodata/reser

## Error in file(file, "rt") : cannot open the connection
## [1] "Skipping due to error - Lake Mead : Area  -  https://www.usbr.gov/uc/water/hydrodata/reservoir_c

## [1] "C:/Rosenberg/Work/USU/Research/ColoradoRiver/RCode/ColoradoRiverCollaborate/AutoReadUSBRData/Ba
## [1] "C:/Rosenberg/Work/USU/Research/ColoradoRiver/RCode/ColoradoRiverCollaborate/AutoReadUSBRData/Res
```

Figure 1: Annual Lake Powell Release Compared to 7.5 and 8.23 Targets

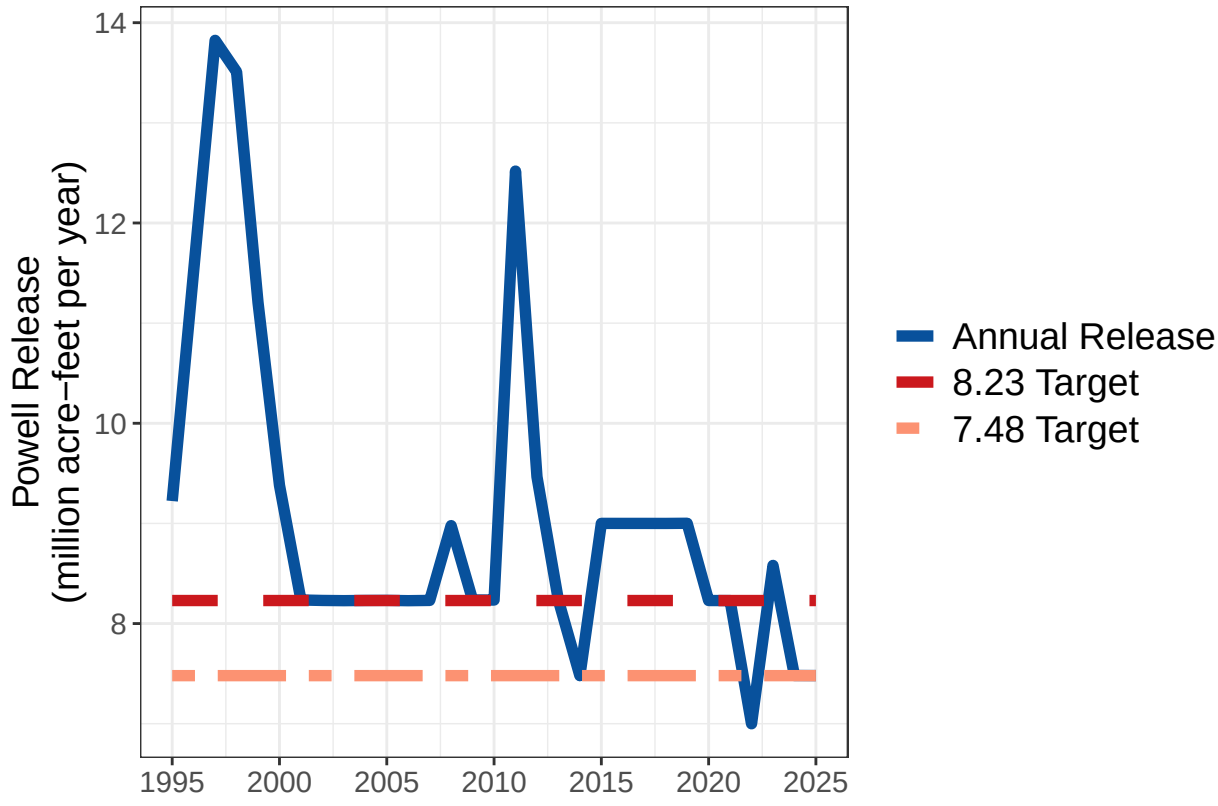


Figure 1: Ten-Year Powell Release Compared to 10-year Targets

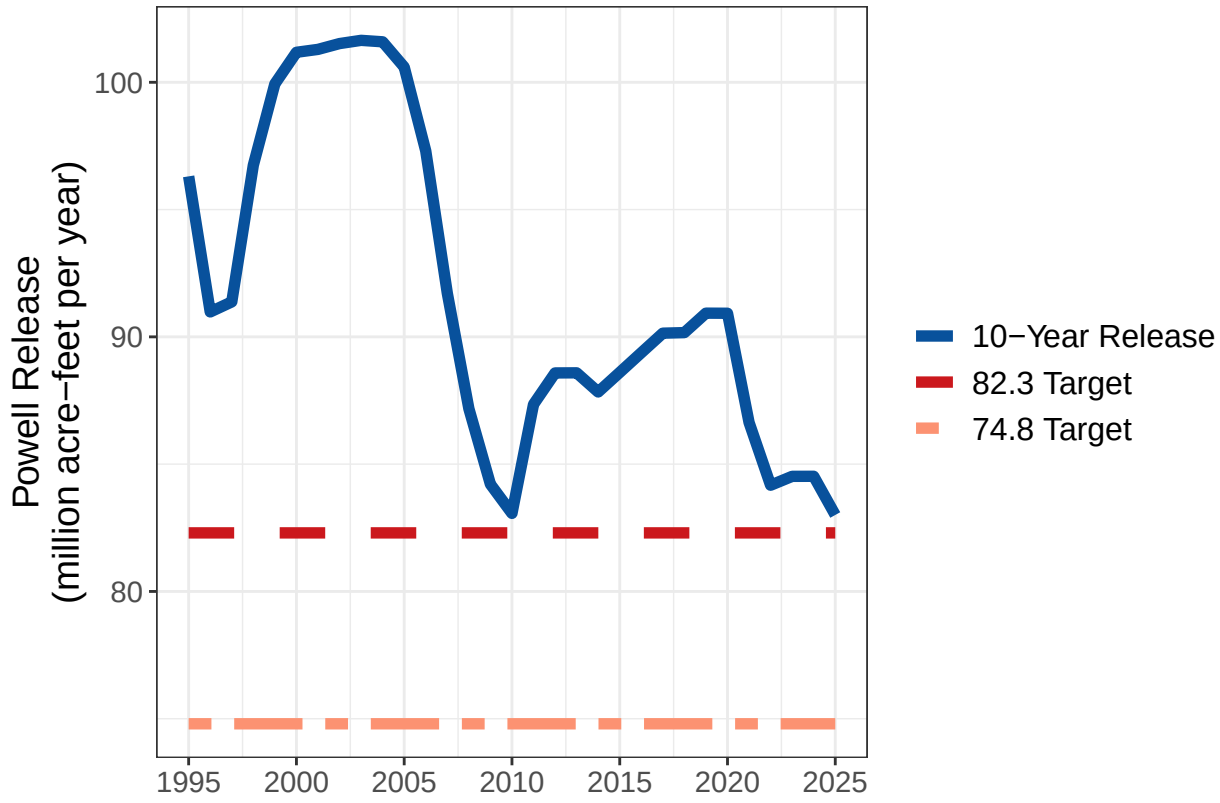


Table 1: Current Storage and Elevations

```
[1] "C:/Rosenberg/Work/USU/Research/ColoradoRiver/RCode/ColoradoRiverCollaborate/HistoricalCurrentReser
[1] "C:/Rosenberg/Work/USU/Research/ColoradoRiver/RCode/ColoradoRiverCollaborate/AutoReadUSBRData/dfRes
[1] "C:/Rosenberg/Work/USU/Research/ColoradoRiver/RCode/ColoradoRiverCollaborate/AutoReadUSBRData/dfRes
[1] "C:/Rosenberg/Work/USU/Research/ColoradoRiver/RCode/ColoradoRiverCollaborate/AutoReadUSBRData/dfRes
[1] "Reservoir Data on: September 27 , 2026"
```

ResName	Storage	Pool Elevation
Lake Powell	5.13	3517.64
Lake Mead	6.83	1037.79

Figure 3: Lake Powell, Lake Mead, and Combined Storage

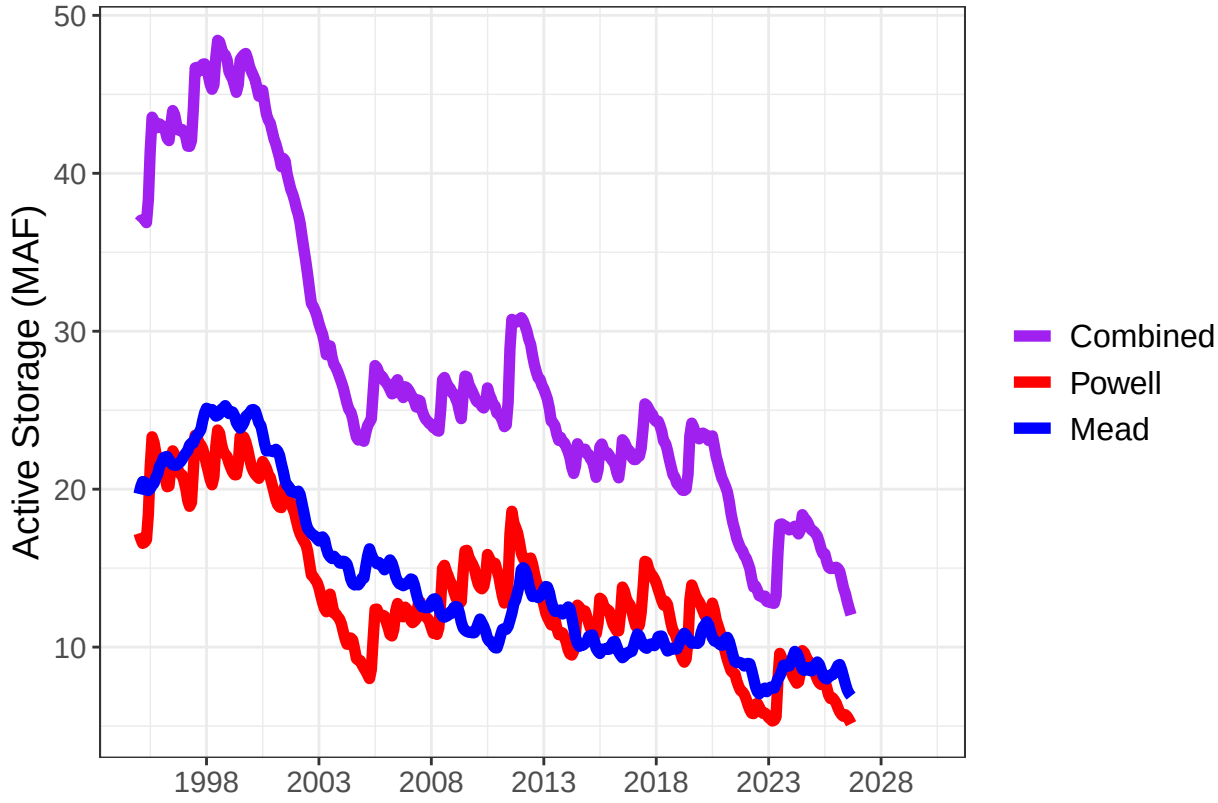


Figure 4: Lake Powell Storage vs Lake Mead Storage

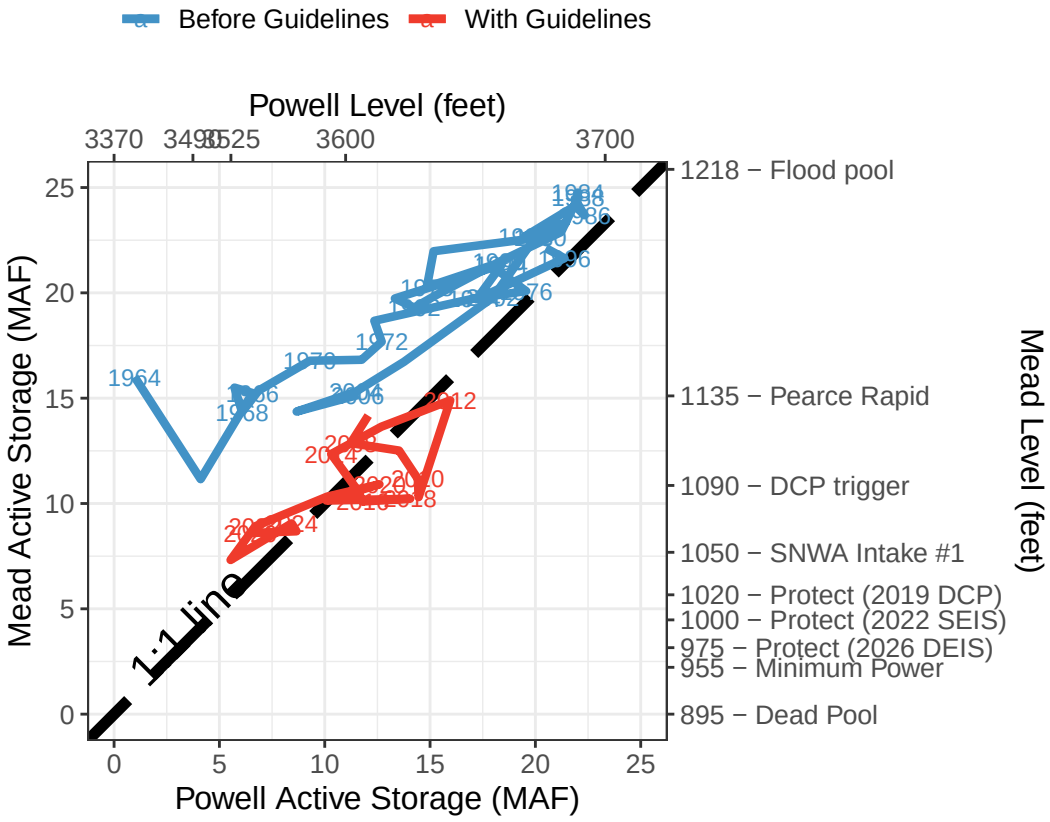


Figure 5: Lake Powell Storage and Elevation over Time.

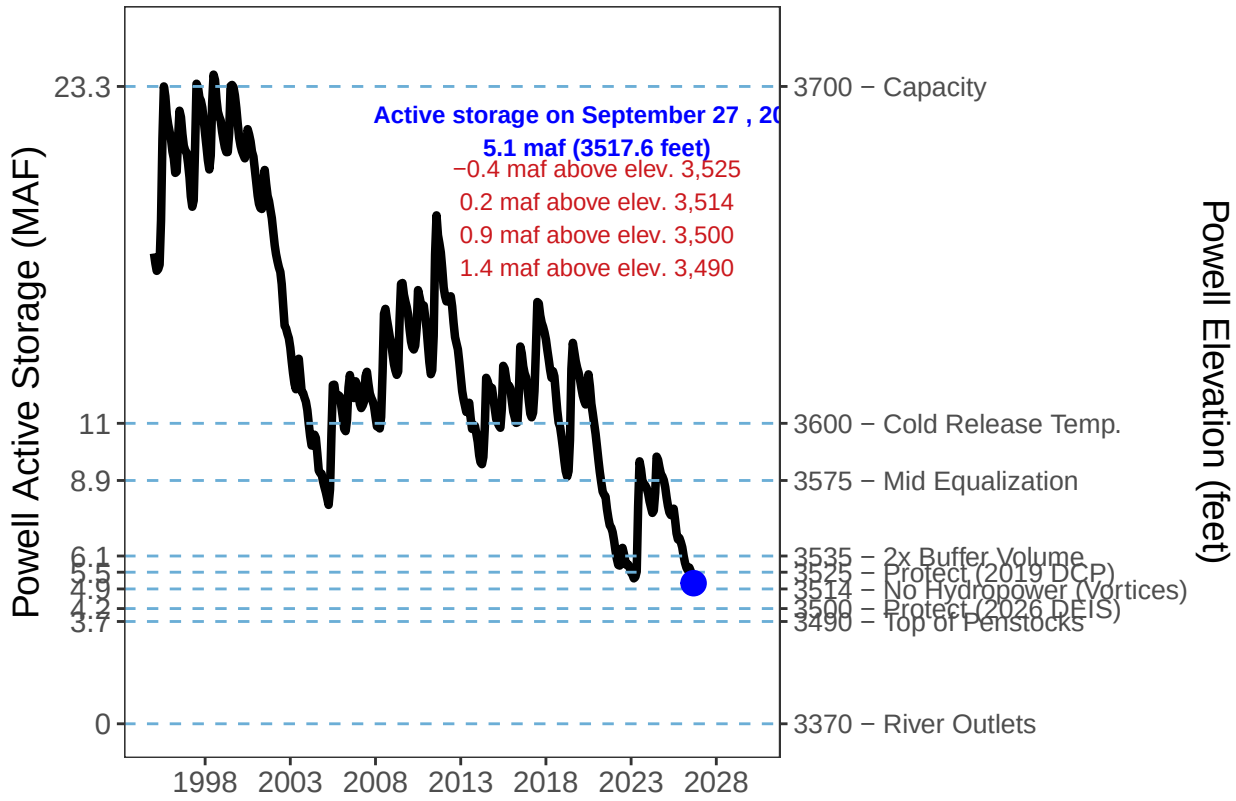
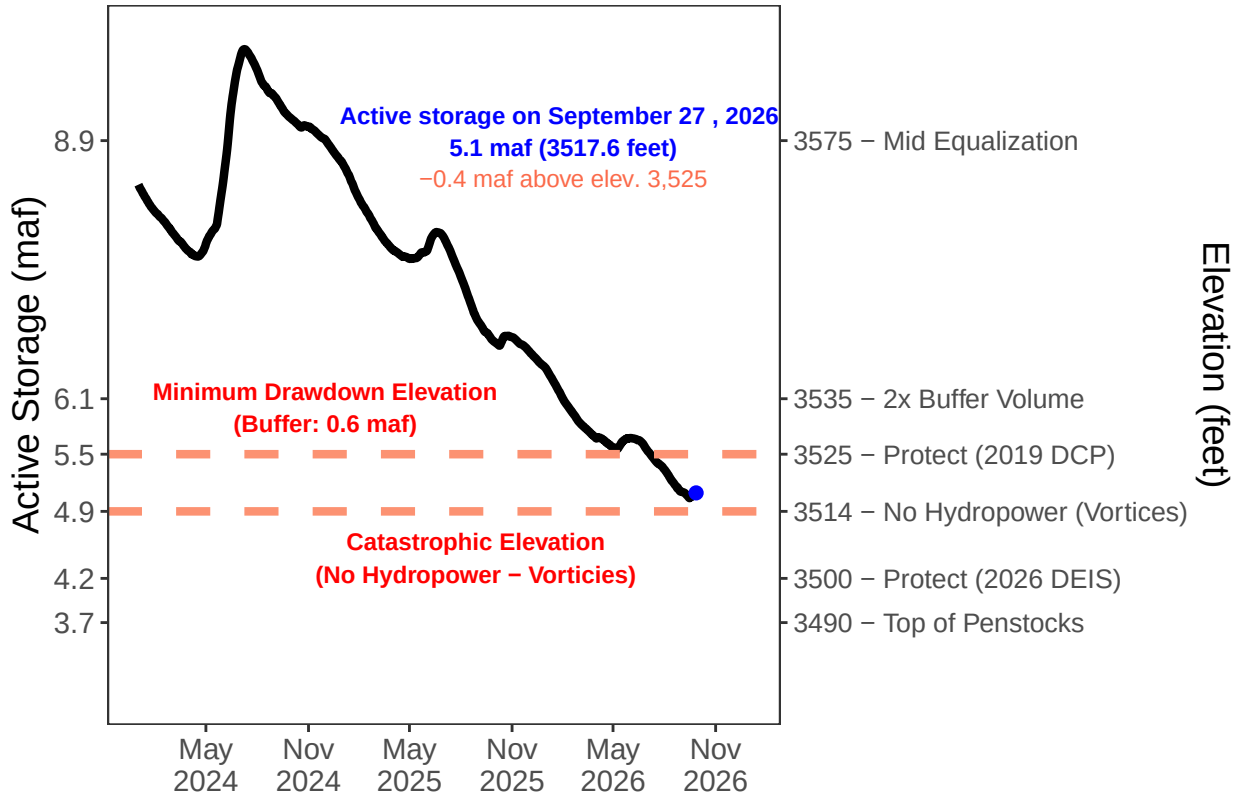


Figure 5B: Lake Powell Minimum Drawdown and Catastrophic Elevations - Daily Data



Recent Lake Powell Decline 5000 acre-feet per day

Recent Lake Powell Decline 35000 acre-feet per week

Table 2. Time Until Powell Elevations

Label	ActiveStorageMAF	DaysUntilVolume	WeeksUntilVolume
3370 - River Outlets	0.00	1016	145
3490 - Top of Penstocks	3.74	275	39
3500 - Protect (2026 DEIS)	4.22	181	26
3514 - No Hydropower (Vortices)	4.93	39	6
3525 - Protect (2019 DCP)	5.54	-82	-12
3535 - 2x Buffer Volume	6.14	-199	-28
3575 - Mid Equalization	8.90	-746	-107
3600 - Cold Release Temp.	10.99	-1160	-166
3700 - Capacity	23.31	-3599	-514

Figure 6: Lake Mead Storage and Elevation over Time.

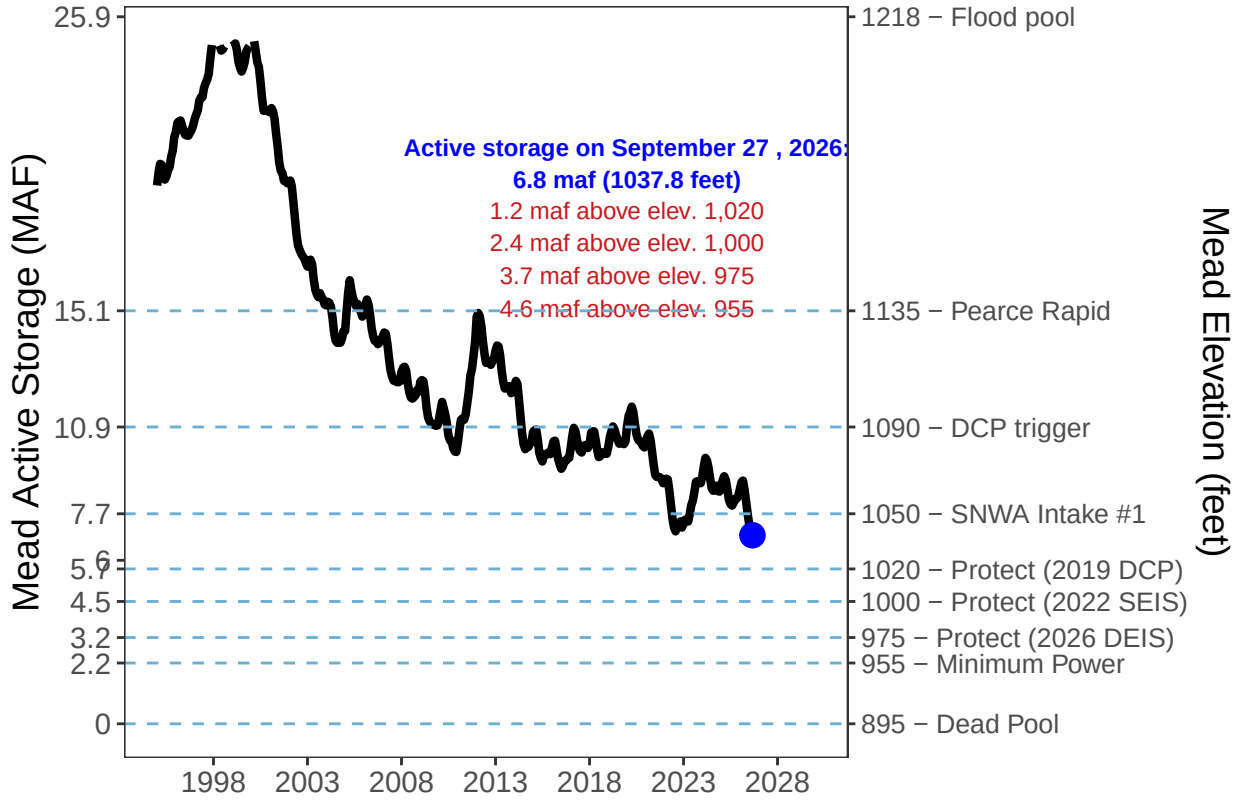


Figure 6B: Lake Powell Daily Inflow and Release Water Year to Date

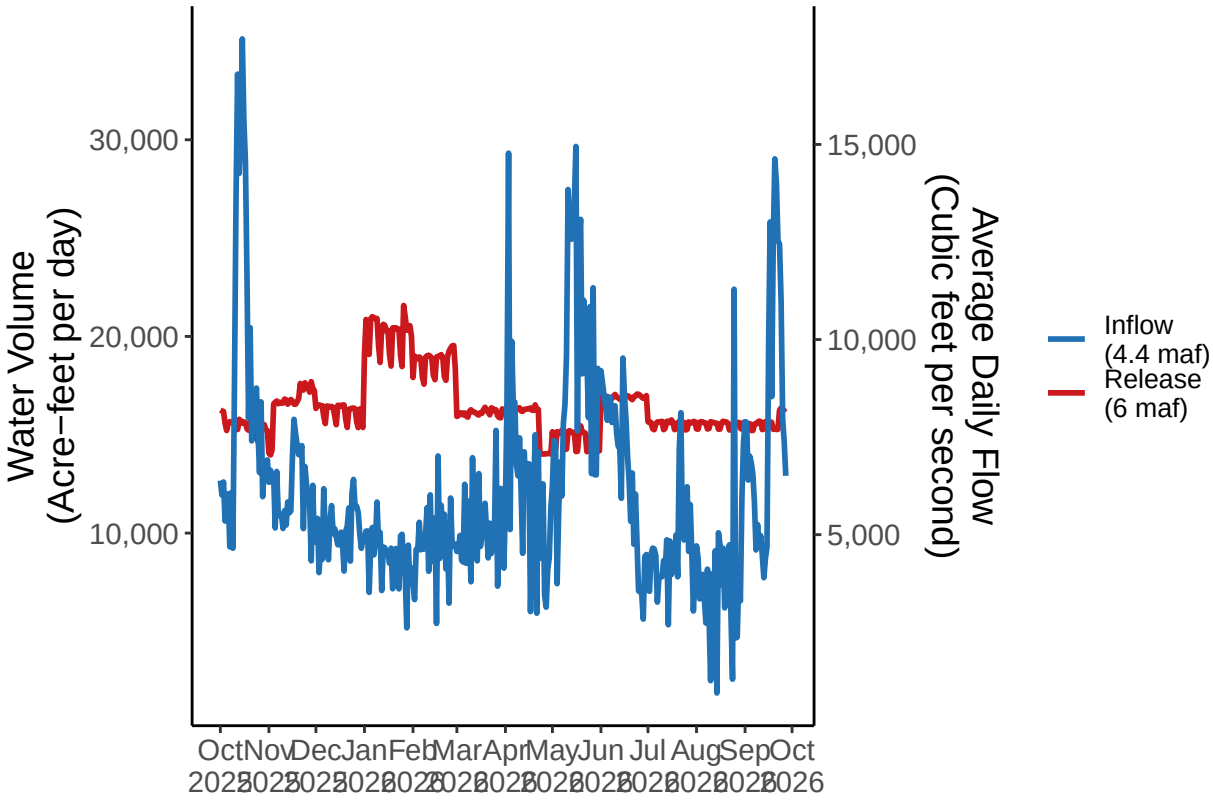


Figure 7: Frequency of Annual Lake Powell Release Volumes.

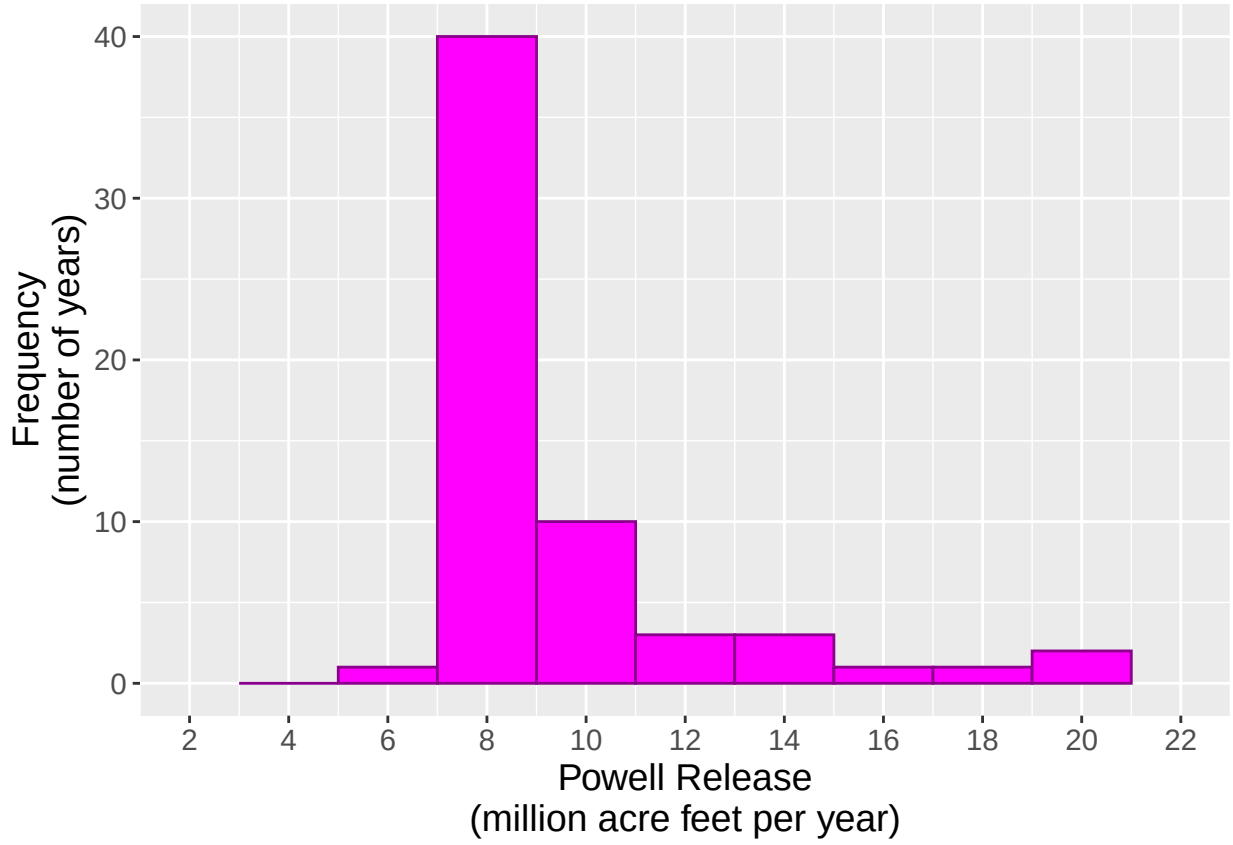


Figure 8: Frequency of Annual Lake Powell Inflow Volumes.

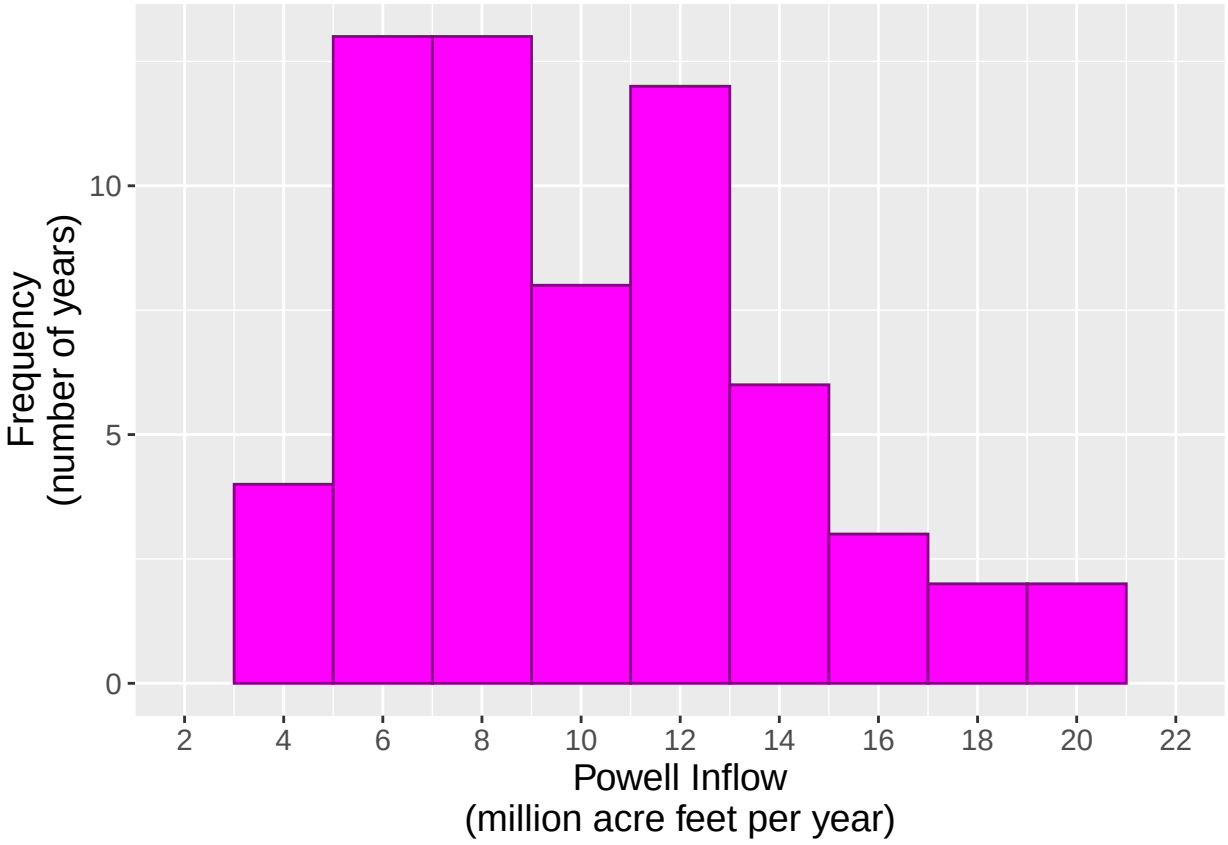


Figure 9: Lake Powell Inflows and Releases.

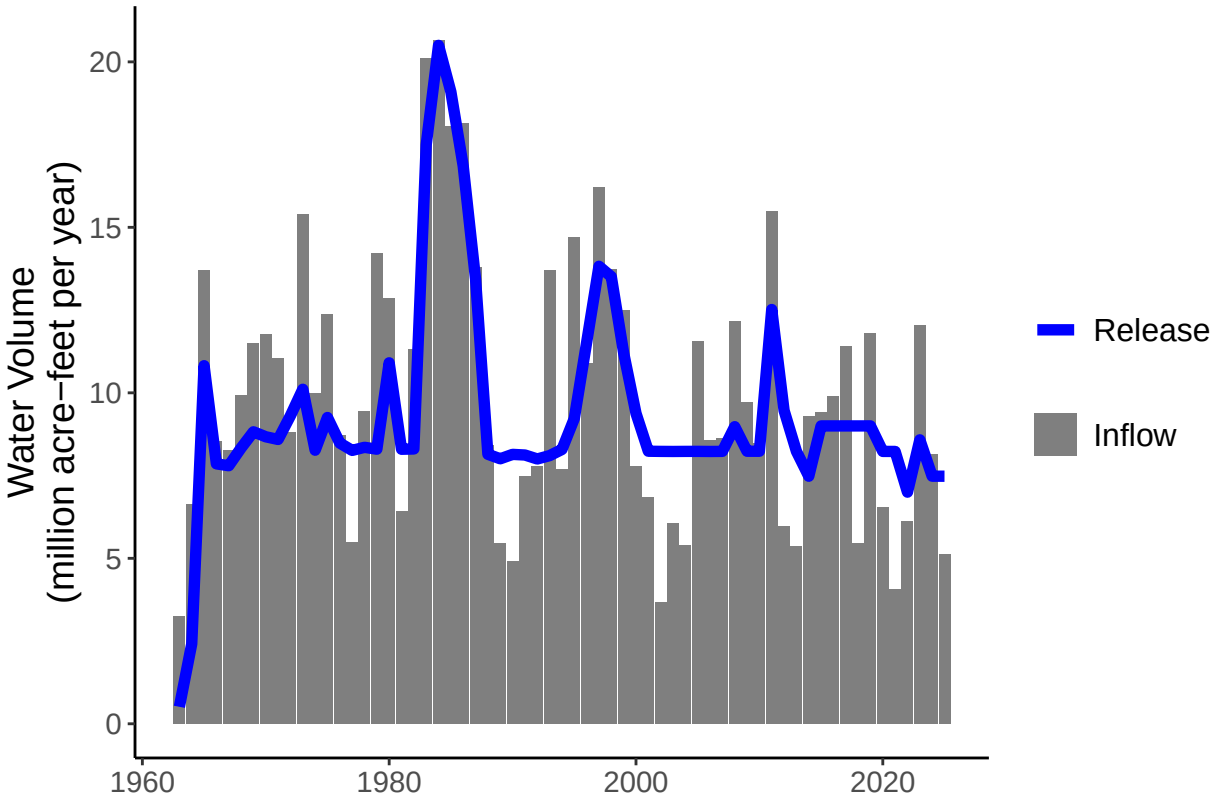


Figure 10: Annual Lake Powell Evaporation.

